

4. Scenario: You are working on a project that involves analyzing the sales performance of a company over the past four quarters. The quarterly sales data is stored in a NumPy array named `sales_data`, where each element represents the sales amount for a specific quarter. Your task is to calculate the total sales for the year and determine the percentage increase in sales from the first quarter to the fourth quarter.

Question: Using NumPy arrays and arithmetic operations calculate the total sales for the year and determine the percentage increase in sales from the first quarter to the fourth quarter?

```
import numpy as np
```

```
# Sales for four quarters
```

```
sales_data = np.array([100000, 120000, 150000, 180000])
```

```
# Calculate total annual sales
```

```
total_sales = np.sum(sales_data)
```

```
# Calculate percentage increase from Q1 to Q4
```

```
percentage_increase = ((sales_data[3] - sales_data[0]) / sales_data[0]) * 100
```

```
print("Total annual sales:", total_sales)
```

```
print("Percentage increase:", percentage_increase, "%")
```

```
===== RESTART: C:/Users/Jannathul firdaus/exp 4 fods.py =====  
Total annual sales: 550000  
Percentage increase: 80.0 %  
>
```

5. Scenario: You are a data analyst working for a car manufacturing company. As part of your analysis, you have a dataset containing information about the fuel efficiency of different car models. The dataset is stored in a NumPy array named `fuel_efficiency`, where each element represents the fuel efficiency (in miles per gallon) of a specific car model. Your task is to calculate the average fuel efficiency and determine the percentage improvement in fuel efficiency between two car models.

Question: How would you use NumPy arrays and arithmetic operations to calculate the average fuel efficiency and determine the percentage improvement in fuel efficiency between two car models?

```
import numpy as np

# Fuel efficiency of three car models (MPG)
fuel_efficiency = np.array([25, 30, 35])

# Calculate average fuel efficiency
average_efficiency = np.mean(fuel_efficiency)

# Compare the first and second car models
old_efficiency = fuel_efficiency[0]
new_efficiency = fuel_efficiency[1]

# Calculate percentage improvement
percentage_improvement = ((new_efficiency - old_efficiency) / old_efficiency) * 100

print("Average fuel efficiency:", average_efficiency, "MPG")
print("Percentage improvement:", percentage_improvement, "%")
```

```
===== RESTART: C:/Users/Jannathul firdaus/exp 5 fods.py =====
Average fuel efficiency: 30.0 MPG
Percentage improvement: 20.0 %
```

6. Scenario: You are a cashier at a grocery store and need to calculate the total cost of a customer's purchase, including applicable discounts and taxes. You have the item prices and quantities in separate lists, and the discount and tax rates are given as percentages. Your task is to calculate the total cost for the customer.

Question: Use arithmetic operations to calculate the total cost of a customer's purchase, including discounts and taxes, given the item prices, quantities, discount rate, and tax rate?

```

# Item prices and quantities
prices = [100, 200, 150]
quantities = [2, 1, 3]

# Discount and tax rates
discount_rate = 10
tax_rate = 5

# Calculate subtotal
subtotal = sum(prices[i] * quantities[i] for i in range(len(prices)))

# Calculate discount
discount = subtotal * discount_rate / 100

# Calculate amount after discount
discounted_total = subtotal - discount

# Calculate tax on discounted amount
tax = discounted_total * tax_rate / 100

# Calculate final cost
total_cost = discounted_total + tax

print("Subtotal:", subtotal)
print("Discount:", discount)
print("Tax:", tax)
print("Total cost:", total_cost)

```

```

===== RESTART: C:/Users/Jannathul firdaus/exp 6 fods.py =====
Subtotal: 850
Discount: 85.0
Tax: 38.25
Total cost: 803.25
>

```